A380X interactive ND

Revise the flight plan on the navigation display with the KCCU cursor, as in the A380 FCOM: DIR TO, INSERT NEXT WPT, DELETE and DATA from a click on the map, with PLACE/DIST entries

Aircraft	A380X
Where to find it	Captain and F/O ND (ARC, PLAN, ROSE-NAV) with the KCCU cursor; MFD FMS > ACTIVE > F-PLN > DIRECT TO
Branch	`core/a380x-interactive-nd`, based on `core/a380x-mfd-fms` (stacked: needs core/surveillance and core/a380x-mfd-fms merged first).
Status	Tested in the sim by the user: OK (before the commit). Unit tests: the DIR TO UTC estimate. PLACE/DIST entries on the ND (added): not sim-tested.
Pictures	No pictures: the ND pages follow the A380 FCOM figures (DSC-31-20-30-90).

1. Interactive ND

Aircraft: A380X | **Where:** Lateral ND, KCCU cursor; MFD DIRECT TO page (UTC) | **Status:** Tested in the sim by the user: OK.

What it is for

A380 FCOM DSC-31-20-30-90: with the KCCU cursor on the lateral ND in ARC, PLAN or ROSE-NAV mode, the flight crew revises the flight plan directly on the ND, without going to the MFD. The FBW ND was not interactive.

What you see

- **Interactive areas:** a blue box around every waypoint, NAVAID or airport under the cursor. A click gives the list of the elements there (P.POS for the aircraft mock-up), then the revision list of the chosen one.
- **Revision list:** the ident (green in the active flight plan, yellow in the temporary one), then: for a flight plan waypoint FROM P.POS DIR TO *, INSERT NEXT WPT, DELETE *; for a database element FROM P.POS DIR TO *; for an airport also DATA; for the aircraft mock-up DIR TO.
- **DIRECT TO page** at the bottom of the ND: the DIRECT TO target field (with a list of the flight plan waypoints, or a KCCU entry), UTC and DIST, the options DIRECT, DIRECT WITH ABEAM, CRS IN, CRS OUT, and the CRS field next to CRS IN / CRS OUT (default course from the flight plan, typed NNN, with M or T for magnetic or true). CANCEL closes the page when there is no temporary flight plan.
- **INSERT DIR TO * / ERASE DIR TO *** buttons either side of the aircraft while the cursor is in the ND; **INSERT TMPY * / ERASE TMPY *** for the other revisions.
- **INSERT NEXT WPT page:** FROM and the revised waypoint (green, or yellow with IN TMPY), its latitude / longitude, and the NEXT WPT field. A click on a blank area of the ND creates a latitude/longitude waypoint.
- **DUPLICATE page** (DIST(NM), LAT/LONG, RETURN, scroll arrows) when a typed ident is not unique, nearest first.
- **DATA** on an airport opens the DATA / AIRPORT page on the MFD of the same side.
- **DIR TO UTC:** the FMS has no predictions of the temporary flight plan, so the UTC at the target is estimated from the distance and the ground speed, on the ND and on the MFD DIRECT TO page. Below 30 kt ground speed it shows --:--.

Limitations and why

Limitation	Why	What to do

No revision on a DIR TO temporary flight plan; no DIRECT TO while the ND shows another temporary flight plan.	FCOM DSC-31-20-30-90 P 1 and P 9.	Insert or erase the temporary flight plan first.
The vertical display is not interactive.	Only the lateral ND is interactive in the FCOM.	None.
Pilot-defined entries (lat/long, typed places) stay as stored waypoints after ERASE.	They are stored waypoints, as on the FBW MFD.	Delete them on the MFD DATA pages if needed.
The UTC is an estimate (distance / ground speed).	The FMS computes no predictions for a temporary flight plan.	Read the UTC on the F-PLN page once the DIR TO is inserted.
Only the interactions listed above work.	The other ND interactions of the FCOM are not implemented yet (the A380X README says so).	Use the MFD pages for the other revisions.

How to use

1. Set the ND to ARC, PLAN or ROSE-NAV (map, not the OANS) and move the KCCU cursor onto the ND.
2. Click a waypoint, NAVAID or airport (or the aircraft symbol); pick it in the list if several are there.
3. Choose the revision: FROM P.POS DIR TO *, INSERT NEXT WPT, DELETE * or DATA.
4. Complete the page at the bottom of the ND (target, option, course, next waypoint) by clicks or KCCU entries.
5. Click **INSERT** (DIR TO * or TMPY *) to activate, or **ERASE** to discard.

How to test

#	Do this	Expected result
1	A380X in flight (or on the ground) with an active flight plan. Captain ND in ARC, range 20 or 40 NM. Move the KCCU cursor over a flight plan waypoint.	A blue box appears around the waypoint.
2	Click it.	Revision list: the ident in green, FROM P.POS DIR TO *, INSERT NEXT WPT, DELETE *.
3	Click FROM P.POS DIR TO *.	DIRECT TO page at the bottom of the ND: the waypoint as target, a yellow temporary flight plan, DIST in NM, UTC (estimated, in flight), DIRECT option selected; INSERT DIR TO * and ERASE DIR TO * either side of the aircraft.
4	Select CRS IN.	A CRS field appears next to it with the default course of the flight plan.
5	Click the CRS field, type a course with the KCCU (for example 090) and ENT.	The course shows; the temporary DIR TO follows it.
6	Click INSERT DIR TO *.	The DIR TO becomes active (green); the page closes.
7	Open the MFD FMS > ACTIVE > F-PLN > DIRECT TO page and select a waypoint.	The UTC shows an estimate (distance / ground speed) instead of dashes.
8	Click the aircraft symbol on the ND, then DIR TO.	DIRECT TO page with an empty target. Open the list (▼) and pick a waypoint, or click the target field and type an ident with the KCCU + ENT.
9	Type an ident that exists several times (a common NAVAID ident).	DUPLICATE page with DIST(NM) and LAT/LONG; pick one, or RETURN.

10	Click ERASE DIR TO *.	The temporary flight plan and the page go.
11	Click a flight plan waypoint > INSERT NEXT WPT, then click a blank area of the map.	INSERT NEXT WPT page with FROM and the waypoint; a latitude/longitude waypoint is created and inserted after it in a yellow temporary flight plan; INSERT TMPY * / ERASE TMPY * buttons.
12	Click INSERT TMPY *.	The new waypoint is in the active flight plan.
13	Click a flight plan waypoint > DELETE *.	Temporary flight plan without that waypoint; INSERT TMPY * keeps the change, ERASE TMPY * discards it.
14	Click an airport > DATA.	The Captain MFD opens the DATA / AIRPORT page of that airport.
15	Repeat one revision on the F/O ND with the F/O KCCU.	Same behaviour; DATA opens on the F/O MFD.

2. PLACE/DIST entries in the ND waypoint fields

> **Aircraft:** A380X | **Where:** ND INSERT NEXT WPT page, NEXT WPT field (KCCU entry) | **Status:** Not sim-tested (added; no test of its own: the PLACE/DIST code is tested on `core/a380x-mfd-fms`).

What it is for

The FMS now accepts PLACE/DIST waypoints (a PDnn waypoint at a distance along the flight plan from a flight plan waypoint, for example `<WPT>/10`): see the PLACE/DIST waypoints feature in the `core/a380x-mfd-fms` manual. The waypoint fields of the interactive ND go through the same shared entry code, which now needs the FMS function that creates the PLACE/DIST waypoint; the ND pages pass it on. Without this, a PLACE/DIST entry typed on the ND would not create the waypoint.

Limitations and why

Limitation	Why	What to do
Only the place of the revised waypoint is accepted (A380 FCOM, INSERT NEXT WPT).	Same rule as on the MFD: another place gives PLACE / WPT DISAGREE, a distance beyond the neighbouring waypoint ENTRY OUT OF RANGE.	-

How to use

1. Click a flight plan waypoint on the ND > INSERT NEXT WPT.
2. Click the NEXT WPT field and type `<that waypoint>/DIST` with the KCCU, then ENT.
3. Check the yellow temporary flight plan and click INSERT TMPY *.

How to test

#	Do this	Expected result
1	A380X in flight with an active flight plan. ND: click a waypoint ahead > INSERT NEXT WPT. Type `<that waypoint>/10` in NEXT WPT + ENT.	A PDnn waypoint 10 NM after it in a yellow temporary flight plan.
2	INSERT TMPY *.	PDnn is in the active flight plan (also on the MFD F-PLN page).

What PR testers should check

- PLACE/DIST entries from the ND fields give the same waypoint as from the MFD.

- Not sim-tested yet.